

A Self-Contained Analysis of the Integer Equation

$$y(x^3 - z^2) = z + 1$$

Abstract

We give a completely elementary proof that the Diophantine equation

$$y(x^3 - z^2) = z + 1$$

has infinitely many integer solutions. More precisely, the entire fibre with $z = -1$ is determined explicitly: it is the union of the two infinite families

$$(x, y, z) = (t, 0, -1) \quad (t \in \mathbb{Z})$$

and

$$(x, y, z) = (1, t, -1) \quad (t \in \mathbb{Z}).$$

For completeness, we also record a full structural characterization of all integer solutions by a single divisibility condition, together with the unique exceptional zero-denominator case.

1 The problem

We study integer triples $(x, y, z) \in \mathbb{Z}^3$ satisfying

$$y(x^3 - z^2) = z + 1. \tag{1}$$

The requested alternatives are either to list all integer solutions or to prove that infinitely many integer solutions exist. The latter is immediate once the fibre $z = -1$ is examined carefully. The point of this note is to present the argument in a fully explicit form, with all elementary facts used in the proof stated and proved.

2 Elementary preliminaries

Lemma 1 (Zero-product property in the integers). *Let $a, b \in \mathbb{Z}$. If $ab = 0$, then $a = 0$ or $b = 0$.*

Proof. Suppose, for contradiction, that $a \neq 0$ and $b \neq 0$. Then $|a| \geq 1$ and $|b| \geq 1$, so

$$|ab| = |a| |b| \geq 1.$$

Hence $ab \neq 0$, contradicting the hypothesis $ab = 0$. Therefore at least one of a, b is zero. $\square$

Lemma 2 (The integer cube equal to one). *Let $x \in \mathbb{Z}$. If $x^3 = 1$, then $x = 1$.*

Proof. By trichotomy, exactly one of the following holds: $x \leq 0$, $x = 1$, or $x \geq 2$. If $x \leq 0$, then $x^3 \leq 0$, so $x^3 \neq 1$. If $x \geq 2$, then $x^3 \geq 2^3 = 8$, so again $x^3 \neq 1$. The only remaining possibility is $x = 1$. $\square$

3 The fibre $z = -1$

Proposition 1 (Exact determination of the fibre $z = -1$). *The integer solutions of (1) with $z = -1$ are exactly*

$$\{(t, 0, -1) : t \in \mathbb{Z}\} \cup \{(1, t, -1) : t \in \mathbb{Z}\}.$$

The union is not disjoint, since both families contain $(1, 0, -1)$.

Proof. Set $z = -1$ in (1). Since $(-1)^2 = 1$ and $(-1) + 1 = 0$, the equation becomes

$$y(x^3 - 1) = 0. \tag{2}$$

By Lemma 1, equation (2) holds if and only if

$$y = 0 \quad \text{or} \quad x^3 - 1 = 0.$$

The second condition is $x^3 = 1$, which by Lemma 2 is equivalent to $x = 1$.

Thus a triple with $z = -1$ solves (1) if and only if either $y = 0$, with x arbitrary, or $x = 1$, with y arbitrary. This is precisely

$$\{(t, 0, -1) : t \in \mathbb{Z}\} \cup \{(1, t, -1) : t \in \mathbb{Z}\}.$$

Conversely, every triple in this union satisfies (2), and therefore satisfies (1) with $z = -1$. $\square$

4 A full structural characterization

The preceding proposition already proves infinitude. The following exact criterion is included to remove any possible ambiguity about the role of the factor $x^3 - z^2$.

Proposition 2 (Divisibility description of all integer solutions). *An integer triple $(x, y, z) \in \mathbb{Z}^3$ satisfies*

$$y(x^3 - z^2) = z + 1$$

if and only if it belongs to exactly one of the following two classes:

- (i) $x^3 - z^2 \neq 0$, the integer $x^3 - z^2$ divides $z + 1$, and

$$y = \frac{z + 1}{x^3 - z^2};$$

- (ii) $(x, z) = (1, -1)$ and y is arbitrary.

Proof. Put

$$D = x^3 - z^2.$$

Then the equation is

$$yD = z + 1. \tag{3}$$

First assume $D \neq 0$. If (x, y, z) is a solution, then (3) says that $z + 1$ is an integer multiple of D , namely the multiple y . Hence $D \mid (z + 1)$ and

$$y = \frac{z + 1}{D} = \frac{z + 1}{x^3 - z^2}.$$

Conversely, if $D \neq 0$, $D \mid (z + 1)$, and $y = (z + 1)/D$, then multiplying both sides by D gives $yD = z + 1$, which is exactly the original equation.

It remains to handle $D = 0$. In that case the left-hand side of (3) is 0, so (3) is equivalent to

$$z + 1 = 0.$$

Thus $z = -1$. Since $D = 0$, we also have

$$x^3 - z^2 = 0.$$

Substituting $z = -1$ gives

$$x^3 - 1 = 0,$$

so $x^3 = 1$, and Lemma 2 gives $x = 1$. Therefore the only solutions with $D = 0$ have $(x, z) = (1, -1)$ and arbitrary y . Conversely, if $(x, z) = (1, -1)$, then $x^3 - z^2 = 1 - 1 = 0$ and $z + 1 = 0$, so

$$y(x^3 - z^2) = y \cdot 0 = 0 = z + 1$$

for every $y \in \mathbb{Z}$.

The two classes are disjoint because class (i) requires $x^3 - z^2 \neq 0$, while class (ii) has $x^3 - z^2 = 0$. $\square$

5 Infinitely many integer solutions

Theorem 1. *The Diophantine equation*

$$y(x^3 - z^2) = z + 1$$

has infinitely many integer solutions.

Proof. By Proposition 1, every triple of the form

$$(x, y, z) = (1, t, -1), \quad t \in \mathbb{Z},$$

is a solution. Directly substituting also verifies this without reference to any previous result:

$$t(1^3 - (-1)^2) = t(1 - 1) = 0 = (-1) + 1.$$

If $t_1, t_2 \in \mathbb{Z}$ and $t_1 \neq t_2$, then

$$(1, t_1, -1) \neq (1, t_2, -1),$$

so these triples are pairwise distinct. Since $\mathbb{Z}$ is infinite, the displayed family contains infinitely many distinct integer solutions. $\square$

Corollary 1. *The equation also has the infinite solution family*

$$(x, y, z) = (t, 0, -1), \quad t \in \mathbb{Z}.$$

Proof. For every $t \in \mathbb{Z}$,

$$0 \cdot (t^3 - (-1)^2) = 0 = (-1) + 1,$$

so $(t, 0, -1)$ is a solution. Distinct values of t give distinct triples. $\square$

Remark 1. No finiteness classification is needed to answer the stated problem, because Theorem 1 proves one of the requested alternatives. Still, Proposition 2 gives an exact necessary-and-sufficient condition for every integer solution: outside the unique exceptional case $(x, z) = (1, -1)$, the value of y is forced by divisibility.